

A Diagnostic Certificate for Dual-Phase Annihilation and Bounded Closure

A Formal (and Intentionally Dry) Note

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Abstract

We formalize a purely diagnostic certificate extracted from a complex phasor field and its induced operator statistics. The certificate consists of (i) a dual-phase symmetry/annihilation lemma, (ii) a closure functional with a singular-cut regularizer, and (iii) a boundedness predicate for the regularized closure (“bounded closure”). All statements herein are internal to the model and are not claims about physical matter/antimatter; the term “annihilation” refers to a symmetric cancellation of phase-alignment statistics.

1 Preliminaries

1.1 Phasor ensemble

Definition 1 (Clause phasors). *Let $C \in \mathbb{N}$ and let*

$$a = (a_1, \dots, a_C) \in \mathbb{C}^C$$

be a complex phasor vector (typically produced by a solver pipeline). Define:

$$r_j := |a_j| \geq 0, \quad \varphi_j := \arg(a_j) \in (-\pi, \pi].$$

Definition 2 (Reference angle and wrapped phase). *Let $\theta \in \mathbb{R}$ be a reference angle. Define the wrapped phase difference*

$$\Delta\varphi_j := \text{wrap}(\varphi_j - \theta) \in (-\pi, \pi],$$

where wrap maps angles to the principal branch.

1.2 Alignment, flip, and Z2-gating

Definition 3 (Raw alignment and flip-set). *Define raw alignment*

$$A_j := \cos(\Delta\varphi_j) \in [-1, 1].$$

Define the flip-set $F := \{j : A_j < 0\}$ and the flipped phases

$$\varphi'_j := \begin{cases} \text{wrap}(\varphi_j + \pi), & j \in F, \\ \varphi_j, & j \notin F, \end{cases} \quad A'_j := \cos(\text{wrap}(\varphi'_j - \theta)) = |A_j|.$$

Definition 4 (Z2-invariant gate). *Define the gate $G_j := (A'_j)^2 = A_j^2 \in [0, 1]$.*

1.3 Empirical statistics used by the certificate

Definition 5 (Empirical mean and spread). *For a real-valued sequence $x = (x_j)_{j=1}^C$, define the empirical mean*

$$\mu(x) := \frac{1}{C} \sum_{j=1}^C x_j$$

and, if needed, the empirical standard deviation

$$\sigma(x) := \sqrt{\frac{1}{C} \sum_{j=1}^C (x_j - \mu(x))^2}.$$

Definition 6 (Key diagnostics). *Assume the pipeline provides the following scalars extracted from a, θ :*

- **Raw alignment mean:** $m_{\text{raw}} := \mu(A)$.
- **Flip fraction:** $f := \frac{1}{C} \#\{j : A_j < 0\} \in [0, 1]$.
- **Phase spread:** $s := \text{Spread}(\{\Delta\varphi_j\}) \in [0, \pi]$ (any fixed, documented spread functional).
- **Bimodality ratio:** $B \in [1, \infty)$ (any fixed, documented bimodality proxy; larger means more two-lobed).

No theorem below depends on the particular implementation of s and B beyond their monotonic semantics.

2 Duality symmetry and “annihilation” (model-internal)

2.1 Symmetry balance

Definition 7 (Symmetry balance). *Define the symmetry balance functional*

$$S(f) := 1 - 2 \left| f - \frac{1}{2} \right| \in [0, 1].$$

Hence $S(f) = 1$ iff $f = \frac{1}{2}$ and $S(f) = 0$ iff $f \in \{0, 1\}$.

Lemma 1 (Elementary properties of S). *$S : [0, 1] \rightarrow [0, 1]$ is continuous, piecewise-linear, maximized at $f = \frac{1}{2}$, and symmetric under $f \mapsto 1 - f$.*

Proof. Immediate from the closed-form definition. □

2.2 Annihilation predicate

Definition 8 (Annihilation predicate). *Fix thresholds (τ_m, τ_S, τ_B) with*

$$\tau_m > 0, \quad \tau_S \in (0, 1), \quad \tau_B > 1.$$

Define

$$\text{ANN}(m_{\text{raw}}, f, B) := \mathbf{1} \left(|m_{\text{raw}}| < \tau_m \wedge S(f) > \tau_S \wedge B > \tau_B \right).$$

Remark 1. In the concrete implementation you described, a typical choice is

$$\tau_m = 10^{-3}, \quad \tau_S = 0.999, \quad \tau_B = 10^6,$$

chosen to match the observed finite-sample behavior of the diagnostics.

Lemma 2 (Interpretation of ANN). *If $\text{ANN} = 1$, then (i) the raw alignment mean is near zero (balanced signs), (ii) the flip fraction is near one-half (duality symmetry), and (iii) bimodality is extreme (two-lobed regime). This is a diagnostic statement; it does not alter any solver decision.*

Proof. Each clause is a direct restatement of the three threshold inequalities. $\square$

3 A “spectrum that sings” index (purely diagnostic)

Definition 9 (Song index). *Given (S, B, s) define*

$$\text{Song}(S, B, s) := S \cdot \frac{\log_{10}(\max\{1, B\})}{\varepsilon + s}, \quad \varepsilon > 0.$$

Remark 2. *The parameter ε is a numerical guard (e.g. $\varepsilon = 10^{-6}$). The user’s “ $\epsilon = 1$ ” corresponds to a deliberate coarse regularization:*

$$\text{Song}_{\epsilon=1} = S \cdot \frac{\log_{10}(\max\{1, B\})}{1 + s}.$$

Proposition 1 (Monotonicity). *For fixed $\varepsilon > 0$:*

1. *Song is non-decreasing in S and in B .*
2. *Song is strictly decreasing in s whenever $S > 0$ and $B > 1$.*

Proof. (1) is immediate since $\log_{10}(\max\{1, B\})$ is non-decreasing in B and the formula is linear in S . (2) follows since $(\varepsilon + s)^{-1}$ is strictly decreasing in s . $\square$

4 Closure and singular-cut regularization

4.1 Closure functional

Definition 10 (Closure functional). *Let $Z_{\text{lock}} \in \mathbb{C}^{T \times C}$ be a locked complex field on a discrete time index $t \in \{1, \dots, T\}$. Assume the pipeline defines a closure integral*

$$\Theta := \text{Closure}(Z_{\text{lock}}) \in \mathbb{R},$$

interpretable as a holonomy-like aggregate of phases along lock loops.

Remark 3. *No theorem below requires the internal details of Closure; only that it is a deterministic scalar functional of Z_{lock} .*

4.2 Refined time

Definition 11 (Refined time). *Given a scale $\lambda > 0$, define refined time*

$$\tau_{\text{ref}} := T + \lambda \Theta.$$

4.3 Singular-cut regularizer (closure-fuse)

Definition 12 (Singular-cut operator on phases). *Let $\alpha = (\alpha_1, \dots, \alpha_C) \in (-\pi, \pi]^C$ be the clause phase angles (e.g. $\alpha_j = \arg(a_j)$). Let $g : (-\pi, \pi] \rightarrow \mathbb{R}_{\geq 0}$ be a measurable “gradient proxy” (implementation-defined). For a chosen percentile $p \in (0, 100)$, let q_p be the empirical p -th percentile of $(g(\alpha_j))_{j=1}^C$.*

Define the singular set

$$\mathcal{S} := \{j : g(\alpha_j) \geq q_p\}.$$

Define a regulated phase vector α^{reg} by applying a cut/soften map

$$\alpha_j^{\text{reg}} := \begin{cases} \text{Cut}(\alpha_j; \delta), & j \in \mathcal{S}, \\ \alpha_j, & j \notin \mathcal{S}, \end{cases}$$

where $\delta > 0$ is a small regularization magnitude and Cut is any deterministic map that reduces $g(\alpha_j)$.

Definition 13 (Fuse diagnostics). *Define the summary:*

$$n_{\text{sing}} := \#\mathcal{S}, \quad \text{frac}_{\text{sing}} := \frac{n_{\text{sing}}}{C}, \quad g_{\text{max}} := \max_j g(\alpha_j).$$

5 Bounded closure (what it should mean, mathematically)

5.1 Boundedness predicate

Definition 14 (Bounded closure predicate). *Fix a bound $G_\star > 0$ and a fraction cap $\rho_\star \in (0, 1)$. We say the closure-fuse step is **bounded** if*

$$g_{\text{max}} \leq G_\star \quad \text{and} \quad \text{frac}_{\text{sing}} \leq \rho_\star.$$

Denote this by

$$\text{BND} := \mathbf{1}(g_{\text{max}} \leq G_\star \wedge \text{frac}_{\text{sing}} \leq \rho_\star).$$

Remark 4. *This definition is intentionally interface-level: it certifies that the regularizer did not produce runaway gradients and did not mark an excessively large portion of clauses as singular.*

5.2 A sufficient condition that matches your printout

Your printout reports e.g. `max_grad=3.142` and `frac=0` and a boolean `bounded=False`. That suggests the implementation uses an additional criterion beyond $(g_{\text{max}}, \text{frac}_{\text{sing}})$, most likely of the form

$$\text{bounded} := (g_{\text{max}} \leq G_\star) \wedge (\text{frac}_{\text{sing}} \leq \rho_\star) \wedge (\text{extra condition}).$$

A robust academic version is:

Definition 15 (Strict bounded closure predicate). *Fix thresholds $G_\star > 0$, $\rho_\star \in (0, 1)$, and a Lipschitz-like cap $L_\star > 0$ on increment. Let $\Delta\alpha := \alpha^{\text{reg}} - \alpha$ (wrapped componentwise). Define the mean regulation magnitude*

$$\Delta_{\text{mean}} := \frac{1}{C} \sum_{j=1}^C |\Delta\alpha_j|.$$

Then define

$$\text{BND}_{\text{strict}} := \mathbf{1}(g_{\text{max}} \leq G_\star \wedge \text{frac}_{\text{sing}} \leq \rho_\star \wedge \Delta_{\text{mean}} \leq L_\star).$$

Remark 5. *This explains a situation where `frac=0` yet `bounded=False`: the regularizer might still report a nontrivial $g_{\max}$ or a nontrivial Δ_{mean} due to global phase curvature or a definition of g that does not vanish even when no points are flagged as singular.*

6 The combined certificate

Definition 16 (Certificate record). *A certificate consists of the tuple*

$$\mathcal{C} := \left(m_{\text{raw}}, f, B, s, S(f), \text{ANN}, \text{Song}, \Theta, \tau_{\text{ref}}, n_{\text{sing}}, \text{frac}_{\text{sing}}, g_{\max}, \text{BND} \right),$$

stored as a diagnostic payload (e.g. in a JSON witness).

Theorem 1 (Separation of concerns: non-interference). *If $\mathcal{C}$ is computed after the solver’s selection of (a, θ) and is not queried by the solver’s control flow, then $\mathcal{C}$ has no effect on the solver output (assignment, score, UNSAT count) and is purely annotative.*

Proof. By construction: the solver output is a function of upstream computations only. If the certificate is evaluated downstream and does not feed back into any upstream parameter, it cannot alter the output. $\square$

7 Implementation mapping (minimal and explicit)

Remark 6 (Where the song index belongs). *Let `op_diag` be the selected-branch diagnostics dictionary containing: `frac_flipped`, `align_raw_mean`, `bimodal_ratio_raw`, `phase_spread`. Then the purely diagnostic additions are:*

$$\begin{aligned} S &:= 1 - 2| \text{frac_flipped} - 1/2 |, \\ \text{Song} &:= S \cdot \frac{\log_{10}(\max\{1, \text{bimodal_ratio_raw}\})}{\varepsilon + \text{phase_spread}}. \end{aligned}$$

This must be stored under `witness["annihilation"]["song_index"]` (or analogous) and printed only.

Remark 7 (Where bounded closure belongs). *Let `fuse_diag` contain at least: `max_grad`, `frac`, `n_singular`. Then define:*

$$\text{bounded} := (\text{max_grad} \leq G_{\star}) \wedge (\text{frac} \leq \rho_{\star}) \wedge (\text{optional strict term}).$$

Store it under `cert_data["closure"]["fuse"]["bounded"]` (or equivalent).

8 Final remark (still dry)

Remark 8. *All terms “duality”, “annihilation”, and “singing spectrum” are model-internal names for specific scalar diagnostics. They are well-defined as functions of (a, θ) and closure-fuse summaries, and can be reproduced deterministically given the same upstream data.*